

Content Calendar Intelligence System for Graphura India Private Limited

1] Executive Summary

This project was developed to design a data-driven content intelligence system for LinkedIn and Instagram. The objective was to analyze historical post-performance and build a predictive model that recommends optimal content strategies to maximize engagement, organic reach, and lead generation.

Using data analytics and machine learning, the system identifies high-performing content patterns and predicts future post-performance categories (High, Medium, Low). The final outcome is a 30-day intelligent content calendar designed to improve brand visibility and engagement consistency.

2] Business Problem

Graphura publishes content across LinkedIn and Instagram but lacked a structured, data-driven framework to answer:

- What type of content performs best?
- Which platform delivers higher engagement?
- What day and time maximize visibility?
- Which hashtag strategies improve reach?
- Can future post performance be predicted?

Without predictive insight, content planning relied on intuition rather than measurable performance indicators.

3] Project Objectives

1. Analyze historical post performance.
2. Identify key engagement drivers.
3. Build a multi-class classification model to predict post success.
4. Develop a recommendation engine.
5. Generate a 30-day optimized content calendar.

4] Methodology

Part 1: Data Analytics

- Collected 176 posts (LinkedIn + Instagram)
- Calculated Engagement Rate:

$$(\text{Likes} + \text{Comments} + \text{Shares}) / \text{Impressions} \times 100$$

- Built dashboard including:
 - Best Performing Platform
 - Best Performing Post Type
 - Best Posting Day
 - Top Hashtags
 - Engagement Rate comparison

Part 2: Machine Learning

- **Target Variable** : Post Performance categorized into: High, Medium, Low (based on engagement rate thresholds)
- **Features Used**
 - Platform
 - Post Type
 - Day of Week
 - Hashtag Count

- Interaction Features
- **Models Trained**
 - Logistic Regression
 - Decision Tree
 - Random Forest
- **Model Evaluation**
 - Stratified Train-Test Split
 - Cross Validation
 - Accuracy Score

Final model selected based on highest cross-validation accuracy.

5] Key Insights from Data

A. Instagram

- Reels generate the highest engagement compared to carousels and static images.
- Moderate-length reels outperform longer videos.
- Hashtag quantity does not directly increase engagement; optimal performance lies within a moderate hashtag range.
- Engagement spikes were observed in specific months (e.g., September), indicating campaign-driven or seasonal impact.
- Mid-week posting (Tuesday–Thursday) tends to perform stronger, while weekends show comparatively lower engagement.

B. LinkedIn

- Document-based posts significantly outperform images and videos.
- Structured, value-driven content generates higher engagement than generic updates.

- Top-performing posts focus on marketing insights, brand growth, and actionable advice.
- Engagement is strongest on Mondays and Saturdays, while Wednesday consistently underperforms.
- Over 55% of posts fall into low-performance category, suggesting opportunity for strategic improvement.

6] Predictive Model Results

- Logistic Regression Accuracy: 0.91 (Selected Model)
- Random Forest Accuracy: 0.91
- Decision Tree Accuracy: 0.88

Logistic Regression demonstrated superior ability to capture non-linear relationships between platform, content type, and Performance variables.

7] Content Recommendation Framework

Based on model predictions, the system recommends:

- Preferred Platform: Instagram for visual and carousel content
- Optimal Post Type: Carousel + Educational Reel Content
- Hashtag Strategy: 5–8 niche and branded hashtags
- Best Posting Days: Tuesday, Monday, Saturday

This transforms content planning from reactive to predictive.

8] Business Impact

- A. Higher Engagement:** Predictive targeting increases probability of High-performing posts, leading to higher engagement rates.
- B. Improved Brand Authority:** Data-driven content enhances credibility and positions Graphura as a structured, analytics-driven organization.
- C. Better Organic Reach:** Optimized hashtags and timing increase algorithm visibility on LinkedIn and Instagram.
- D. Increased Inquiries:** High-performing educational and internship-related posts increase:
- Internship applications
 - Service inquiries
 - Direct messages
 - Website visits
- E. Repeatable Strategy Framework:** The system creates a reusable framework:

Data → Analysis → Prediction → Recommendation → Calendar → Performance Monitoring

This allows continuous improvement over time.

9] Conclusion

The Content Calendar Intelligence System successfully integrates data analytics and machine learning to create a predictive content optimization framework.

This project demonstrates how AI-driven insights can directly translate into measurable marketing performance improvements.

By implementing this system consistently, Graphura can expect:

- Increased engagement consistency
- Stronger brand presence
- Improved organic growth
- More inbound leads

10] Future Enhancements

- Sentiment Analysis of captions
- NLP-based hashtag clustering
- Deep learning model integration
- Real-time dashboard with Streamlit
- Automated content scoring system